Kafka Consumer Group Rebalance - Part 2 of 2

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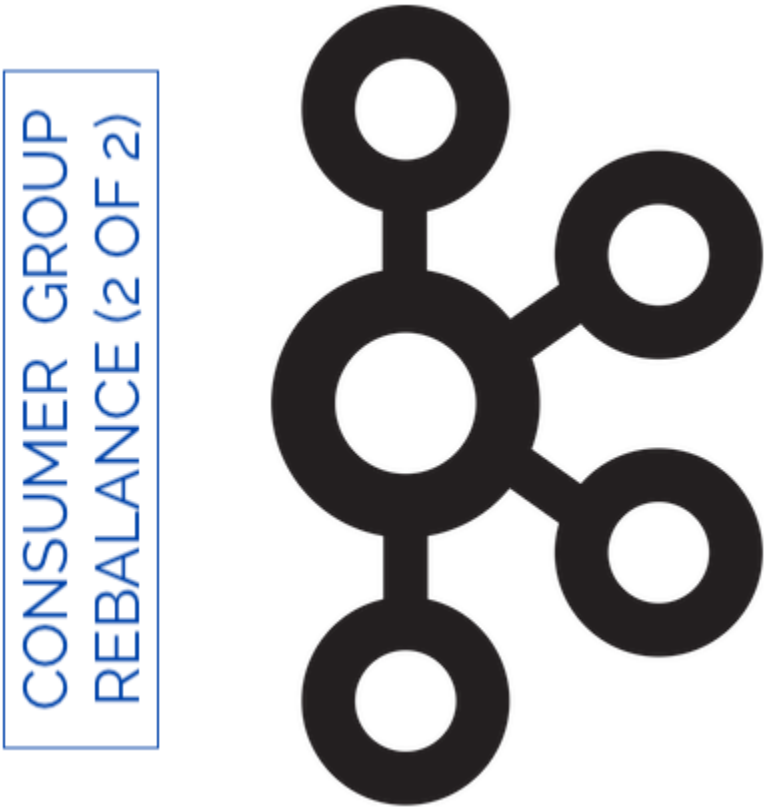

Introduction

This is the second in a two part article on Consumer Group Rebalance. In the first part consumer groups, rebalances, triggers for a rebalance, and the configuration options that impact rebalances were covered.

In this second part of the article the rebalance strategies available, the ability to reduce the number of unnecessary rebalances with Static Group Membership, and the risks to consider with rebalancing are all explored.

Rebalance Strategies

Eager Rebalance

With eager rebalancing (the default), when a consumer group rebalances, all processing by the consumers stops while the topic partitions are reassigned. This means the number of rebalances and their impact on an application is critical to understand as it can have a significant impact on throughput.

The following diagram illustrates the impact on an existing consumer group containing a single consumer when a new consumer joins the group, and the time outs in play. However it is applicable to all three rebalance trigger scenarios, whether the Group Coordinator has received a JoinGroup or a LeaveGroup request from a consumer, or it believes a consumer may have failed.

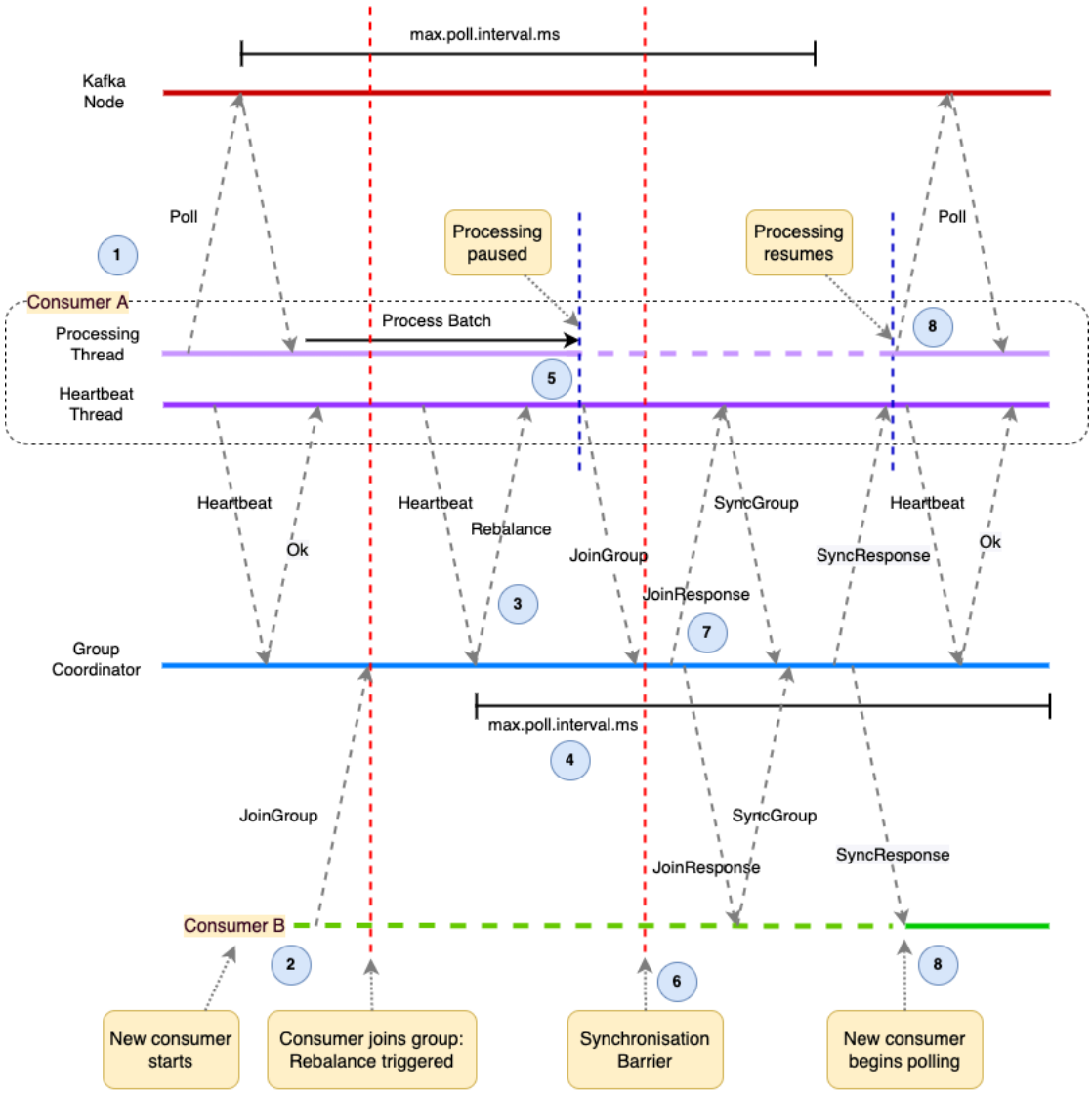

Figure 1: Eager Rebalance

1 Consumer A is polling the topic partition for messages, and heart beating to the Group Coordinator to tell it is healthy. The response to this heartbeat from the Group Coordinator is an acknowledgement of ‘Ok’, and the consumer continues its processing.

2 Consumer B starts and sends a JoinGroup request to the Group Coordinator triggering a rebalance. (Just the heartbeat thread for Consumer B is shown on the diagram for clarity).

3 The Group Coordinator responds to the next heartbeat it receives from its existing consumer, Consumer A, notifying it that a rebalance has begun.

4 Consumer A has until its max.poll.interval.ms to complete processing the messages from its current poll() and respond with its own JoinGroup request.

5 Consumer A’s poll completes and now all message processing is paused. Its next call to poll() triggers the send of a JoinGroup request with the information on the topic subscriptions it is interested in to the Group Coordinator.

6 The Group Coordinator, being aware of all the consumers in the group, knows the point at which all existing topic partitions have been released and can be reassigned. The ‘Synchronisation Barrier’ has been reached as the consumer group has stabilised with two members.

7 The Group Coordinator sends JoinResponses to both consumers. One consumer is selected as leader and calculates the partition assignments. The consumers respond with SyncGroup requests. The group leader’s SyncGroup request contains the computed partition assignments.

8 The Group Coordinator responds with a SyncResponse to each consumer notifying them of their assignments. Consumer A now resumes polling, and Consumer B begins polling.

The consumer group rebalance does not complete until all consumers have accepted their partition assignments. The diagram highlights the pause in processing during rebalance, and as the consumer group increases in size with more members the duration of this pause becomes more significant.

Incremental Rebalance

The larger a consumer group is, the longer it can take for a rebalance to take place. If the impact of eager consumer group rebalances stopping message processing while they are occurring is considered too great, then an Incremental Rebalance strategy could be adopted. (This is also known as Cooperative Rebalance). This time existing consumers that have been notified by the Group Coordinator that a rebalance is underway do not stop processing. Instead rebalancing occurs over two phases. As the consumers receive notification from the Group Coordinator that a rebalance has begun, the following now occurs:

1. The existing consumers send a JoinGroup request to the Group Coordinator, but continue processing messages from their assigned topic partitions.
2. The JoinGroup request contains their topic subscription information and the information on their current assignments.
3. Once the Group Coordinator receives JoinGroup Requests from all existing consumers (or they have timed out), it then sends JoinResponses to the consumers, and assigns a new group leader.
4. The new group leader responds with a SyncGroup request with the intended consumer to partition assignments.
5. The Group Coordinator notifies the consumers that must release partitions in a SyncResponse.
6. Only those consumers that need to revoke partitions actually stop processing in order for those partitions to be reassigned to another consumer.
7. New JoinGroup requests are sent from all the consumers to the Group Coordinator in a second round of the rebalance protocol with information on the partitions they still own and those they have revoked.
8. At this point the group has stabilised and the rebalance has reached the 'Synchronisation Barrier'. Partition assignment can now be completed.

Only those partitions that need to be reassigned are revoked. The other partitions are constantly owned by their consumers with no interruption to consumption of their messages.

The following diagram illustrates the incremental balance in action.

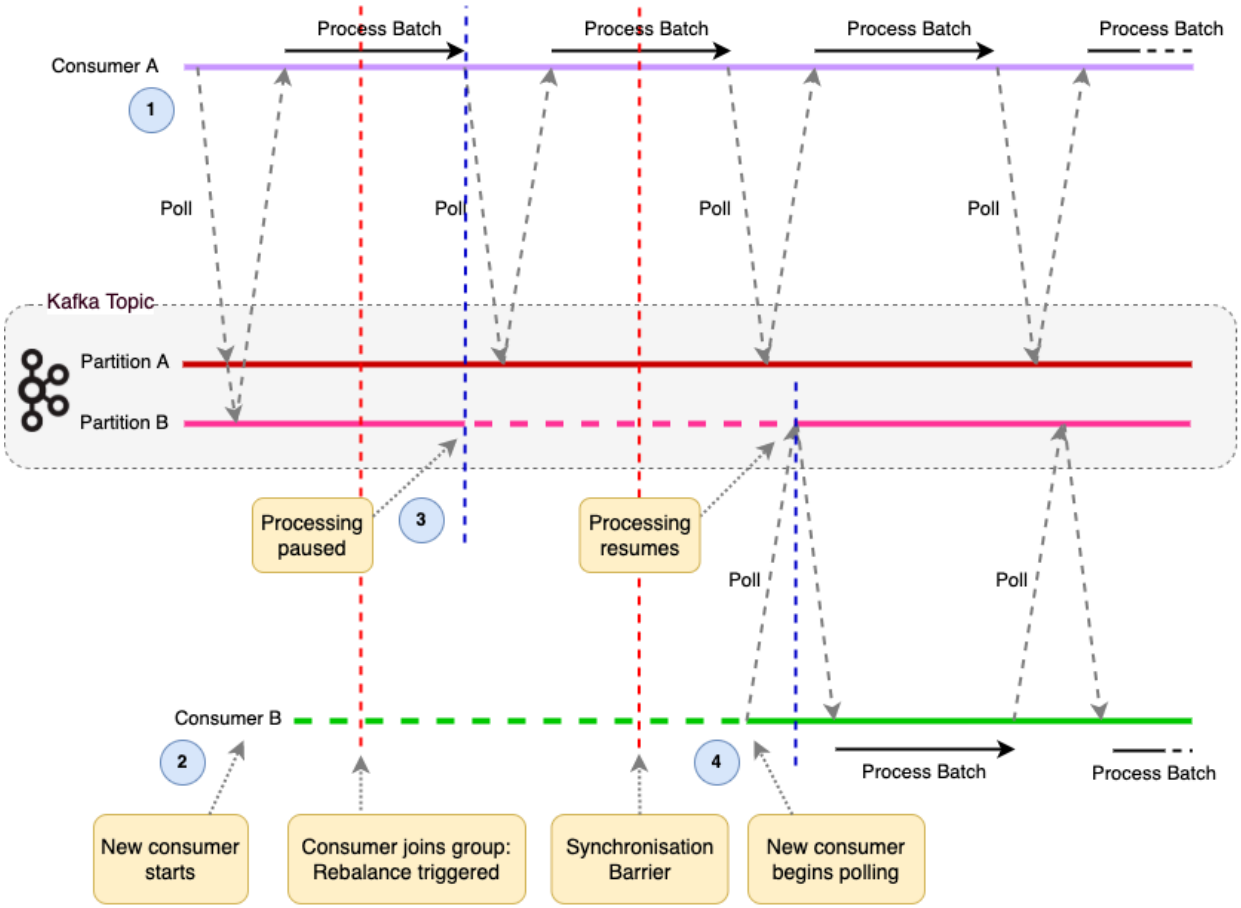

Figure 2: Incremental Rebalance

1 Consumer A is polling from two partitions when a second consumer.

2 Consumer B, starts and joins the group. This triggers the incremental rebalance.

3 Consumer A gives up its assignment of one of its partitions.

4 The partition is reassigned to Consumer B which begins consuming messages from it. Meanwhile Consumer A does not stop processing from the other of the two partitions.

The diagram does not include the complexity around the heartbeating.

Incremental Rebalance takes two rounds of rebalancing to complete, so results in longer overall latency. However the impact of the rebalance is less severe to overall message processing.

Incremental Rebalance is configured by applying a CooperativeStickyAssignor to the consumer's partition.assignment.strategy setting.

Static Group Membership

The number of unnecessary rebalances, and hence the impact of rebalancing on throughput, can be reduced by using Static Group Membership. With the default rebalance protocol when a consumer starts it is assigned a new member.id (which is an internal Id in the Group Coordinator) irrespective of whether it is an existing consumer that has just restarted. Any consumer starting triggers a rebalance, and is assigned a new member.id. With this protocol the consumer cannot be re-identified as the same.

The Static Group Membership protocol introduces the ability to configure a unique group.instance.id on the consumer, marking it as a static member. The Group Coordinator maps this group.instance.id to the internal member.id. If a consumer dies and restarts it will send a JoinGroup request with this id to the Group Coordinator. In the scenario where the consumer shuts down it is not removed from the consumer group until its session times out based on the session.timeout.ms. When the consumer is restarted and rejoins the group, the Group Coordinator checks and finds the group.instance.id matches that of a static member it has registered in the consumer group. It therefore knows it is the same consumer instance and a rebalance is not triggered. The partitions that were assigned to that consumer are reassigned to it and processing of messages from those partitions now resumes. Meanwhile there was no interruption to the processing of messages on partitions assigned to other consumers.

The following diagram demonstrates static group membership. Two consumers belong to the same consumer group and have distinct group.instance.id values assigned. They are polling a partition each from the same topic. Consumer B stops and leaves the group, however a rebalance is not immediately triggered. The consumer rejoins the group before the session.timeout.ms times out and is reassigned its partition, ensuring no rebalance is required.

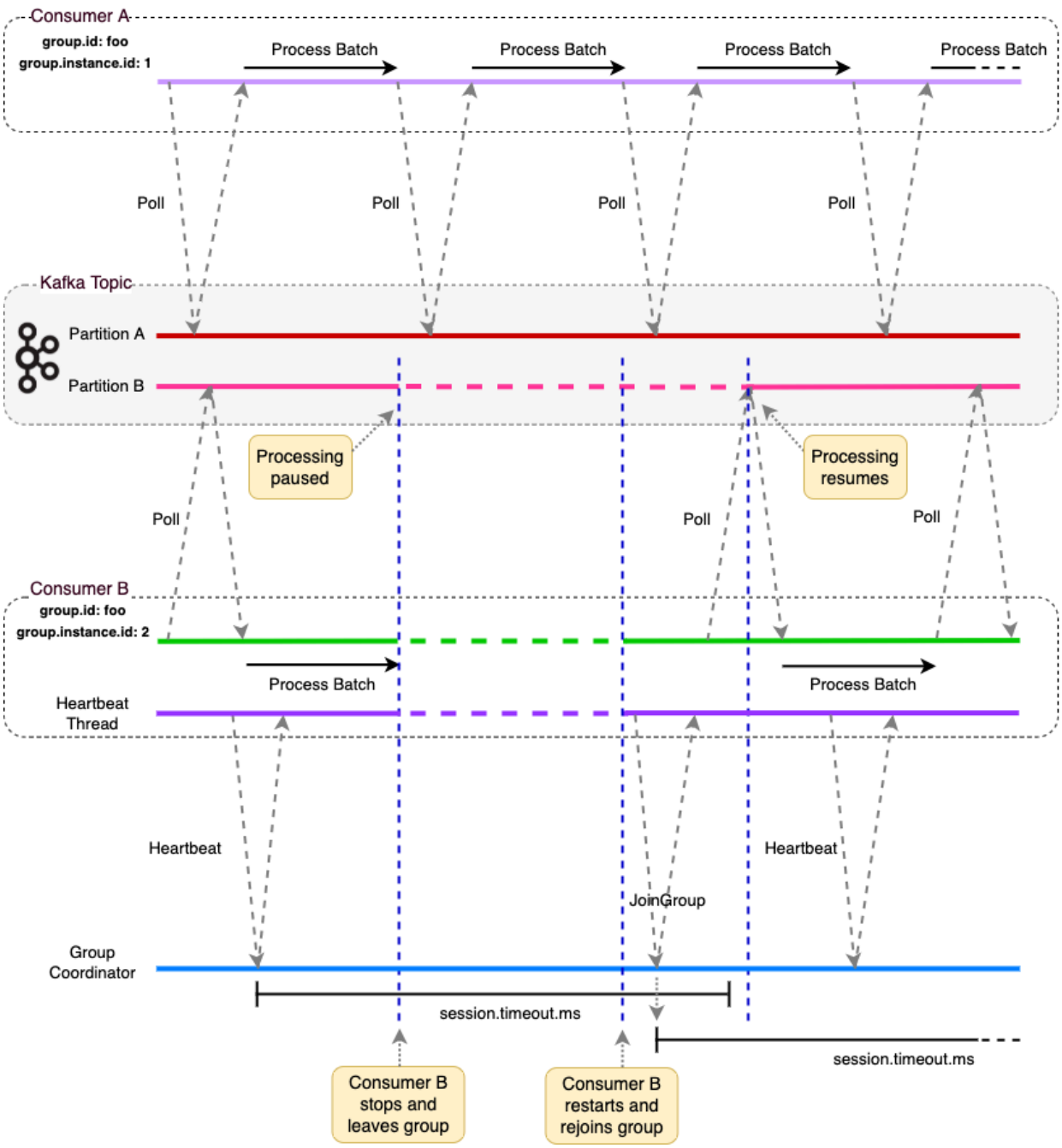

Figure 3: Static Group Membership

For clarity on the diagram the heartbeat thread is only shown for Consumer B.

This feature could be utilised for example by tieing the Id of the Kubernetes pod that an application is running in to the application consumer's group.instance.id. If the pod dies and restarts then the Group Coordinator will recognise the consumer as the group.instance.id will be the same, and the potentially costly rebalance is avoided.

Static Group Membership is of particular interest when maintaining state in the consumer that is otherwise lost, or must be reloaded, following a rebalance. For example, stateful retry allows a consumer to track retries of message batches across polls. The retry count would be lost on a consumer rebalance if the partitions being polled are assigned to other consumers. Stateless vs stateful retry is covered in the article here.

Care needs to be taken when using Static Group Membership as when a consumer has died those partitions it was assigned will not be reassigned until the consumer has timed out. Therefore configuring a longer session.timeout.ms to allow a restarting consumer time to rejoin and avoid triggering a rebalance comes with the risk that a genuinely failed consumer that does not rejoin will leave partitions without a consumer assigned for longer. However configuring a session.timeout.ms that is too short may not allow enough time for a consumer to rejoin before it is removed from the consumer group. With the consumer no longer in the consumer group when it rejoins a rebalance is triggered.

For a consumer with static group membership it does not send a LeaveGroup request when it leaves a group (or indeed fails). Rather it stops heartbeating and remains in the group until the session.timeout.ms has been exceeded and is removed from the group by the Group Coordinator. This timeout should then be configured to be sufficiently long to allow time for the consumer to restart and be reassigned its partitions without the need for a rebalance.

Rebalance Risks

Duplicate Messages

A consumer that has exceeded its time out and is considered failed could still be processing the messages it has polled, and that processing could complete successfully. However its consumer offsets write will be rejected as the consumer group rebalance increments the generation Id, and any writes with the previous generation Id are rejected. Meanwhile a new consumer instance is assigned the topic partitions in a rebalance and this consumes and processes the same messages. It is always important to be aware that the application may receive duplicate messages and it must cater for these, if necessary, as required. More on deduplication patterns here.

Rebalance Storms

Rebalance does not complete until all existing consumers have either rejoined or exceeded the max.poll.interval.ms. If a consumer does indeed exceed the max.poll.interval.ms before it again polls as it is taking longer than expected to process its last batch of messages then when it does complete it will request to rejoin the group, triggering another rebalance. If the cause of the rebalance is for example due to slow responding downstream services that are

affecting all consumers the upshot can be rebalance after rebalance being triggered as consumers are continually evicted and then rejoin, a ‘rebalance storm’. Static Group Membership and Incremental rebalancing can of course assist with this but whatever strategies are in place care must be taken with the rebalance configurations.

Conclusion

Consumer Group rebalance is a critical part of how Kafka manages consumer groups, which itself is an important feature that helps make Kafka a highly scalable distributed messaging. Understanding how rebalance works and how the various consumer configurations affect this is essential in ensuring that throughput is maximised and the system does not suffer from frequent periods where messages are not being processed.

Previous...

For an in-depth look at consumer groups, rebalance, rebalance triggers, and the configuration options around rebalance timeouts, see the first part of this article.

Resources

- Kafka - The Definitive Guide: Real-Time Data and Stream Processing at Scale by Gwen Shapira, Todd Palino, Rajini Sivaram
- Matthias Sax's talk: Everything You Always Wanted to Know About Kafka's Rebalance Protocol But Were Afraid To Ask